

Agentic wealth management platform

A conversational agent that builds a diversified, risk-minimized portfolio for each user, then watches the market on their behalf — flagging news that matters, never acting without permission.

Audience

Pakistani retail investors

Stack

FastAPI • React/Vite • PostgreSQL

Model tier

Free-first (Gemini • Groq)

Status

Architecture locked

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01 Problem & opportunity

Retail investors in Pakistan have capital and intent but no accessible way to build and manage a diversified portfolio across the asset classes available to them.

Existing tools either track a single asset class, assume a sophistication the average user lacks, or are priced for institutions. Meanwhile the data that *would* power a good advisor — PSX prices, SBP rate moves, dividend announcements, commodity trends — is scattered, partly free, and never assembled into one decision-making view. The investor is left reacting to news they may not even know affects their holdings.

- ◆ **Fragmented data.** PSX, T-bills, commodities, and crypto each live behind different sources with different access models.
- ◆ **No personalization.** Generic market news isn't filtered to what a specific person actually holds.
- ◆ **No continuity.** Tools recommend once and stop; nobody watches the portfolio after the money is in.
- ◆ **Local constraints ignored.** Most models use a 0% risk-free rate; in Pakistan, the SBP policy rate is a dominant factor that changes every allocation.

OPPORTUNITY

A single agent that ingests the available free data, reasons over it with real portfolio theory tuned to Pakistan, and stays on as a persistent monitor — doing the analytical work continuously while leaving every decision with the user.

02 Product vision

A conversational advisor that profiles you, proposes a portfolio, and then watches the market for you — like a diligent analyst who only ever recommends, and waits for your call.

The user talks to the agent in plain language. The agent profiles their risk, horizon, and capital; pulls candidate assets; runs portfolio optimization with diversification and risk constraints tuned to the local market; and returns a suggested allocation with a plain-language rationale. Once the user confirms what they actually invested, the agent shifts into a monitoring role: it tracks news against those exact holdings, scores each event for materiality, and flags only what genuinely changes the portfolio's risk or return. When something material hits, the user — not the agent — decides whether to re-optimize.

NORTH-STAR PRINCIPLE

Autonomous analysis, human-approved decisions. The agent does all the watching, matching, and computing on its own. It never moves money or rebalances without the user pressing the button.

03 Users & modes

The product serves two investor postures. The agent classifies which the user is during onboarding and adapts its logic, data cadence, and recommendations accordingly.

DIMENSION	LONG-TERM INVESTOR	SHORT-TERM / ACTIVE
Horizon	Months to years	Days to weeks (swing)
Focus	Diversification, dividends, fundamentals	Momentum, volatility, technicals on daily data
Agent provides	Asset allocation, rebalancing schedule, dividend tracking	Technical indicators, volatility ranking, position sizing
Risk control	Correlation-based weighting, sector limits	Stop-loss suggestions, max-drawdown limits
Data cadence	Daily EOD is sufficient	Daily EOD; intraday is out of free-data scope

HONEST SCOPING NOTE

True intraday day trading needs tick data that isn't freely available for PSX. "Active" mode is deliberately built on daily candles and framed as short-term/swing — an honest scope given the data, rather than implying live tick feeds the system doesn't have.

04 The agent lifecycle

The agent moves through three phases. This lifecycle is the product — each phase has distinct responsibilities and a distinct portfolio state.

PHASE 01

Onboarding

User fills a form / chats.
Agent profiles them, pulls candidate assets, runs optimization, and suggests a portfolio with rationale. State: **draft**.

PHASE 02

Activation

User confirms what they actually invested (may differ from the suggestion). Agent locks the real holdings and begins tracking. State: **confirmed**.

PHASE 03

Monitoring

Background jobs watch news and prices against confirmed holdings, score materiality, and flag. User triggers any re-optimization. State: **tracked**.

WHY STATE MATTERS

The agent monitors the *confirmed* portfolio, not the suggestion. Users rarely buy exactly what's proposed, so the system must store actual positions as the source of truth for all tracking and performance.

05 End-to-end flow

The full path from a new user to an actively monitored portfolio. Read top to bottom; the loop at the base is the continuous monitoring cycle.

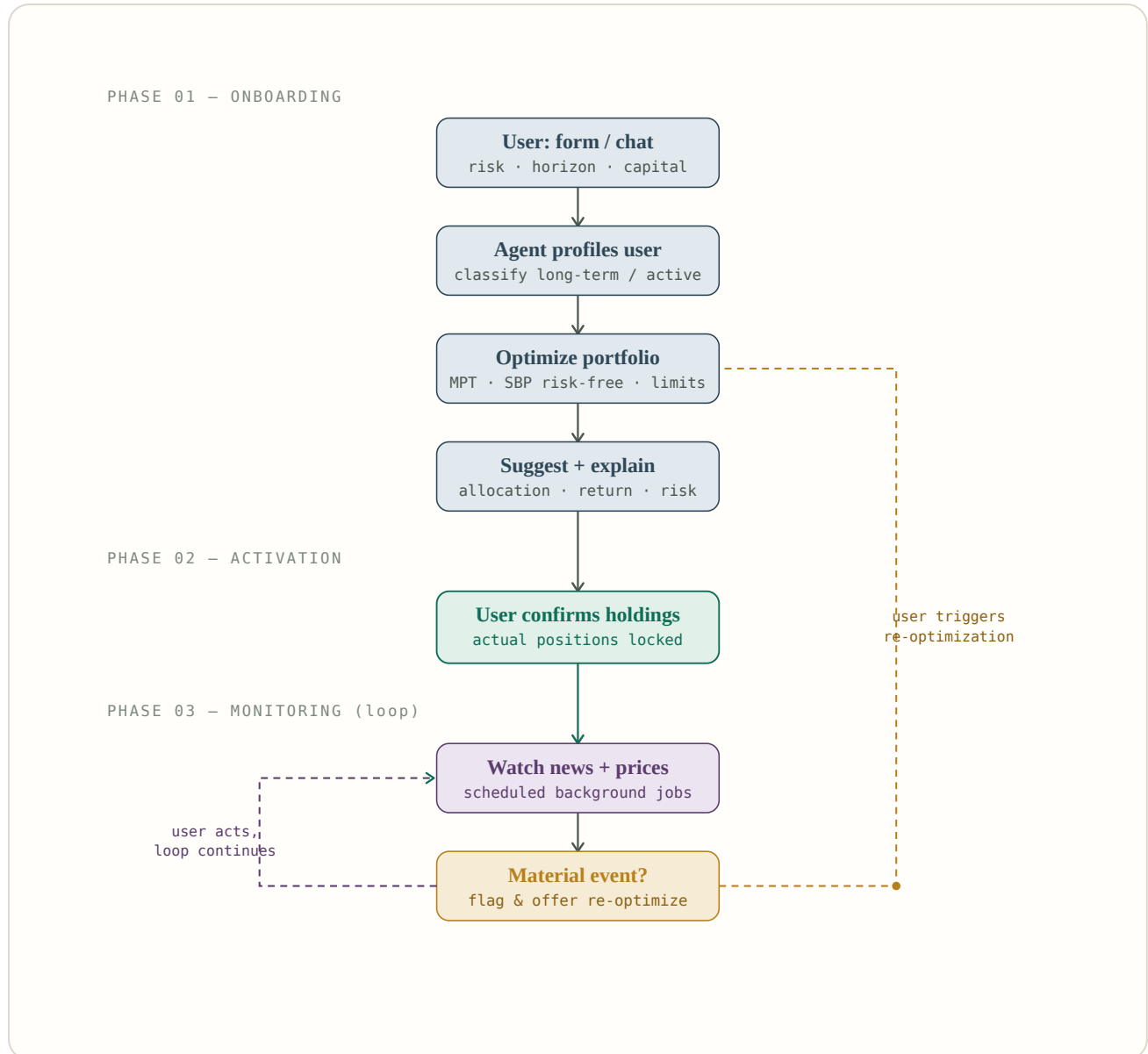


Figure 1 · **End-to-end agent flow** – three phases, with the monitoring loop and the user-triggered re-optimization path

06 News & materiality flow

The monitoring engine's core logic. Raw news is noise; the agent's real job is filtering to what affects *this* user's holdings, then deciding whether it's actionable or merely informational.

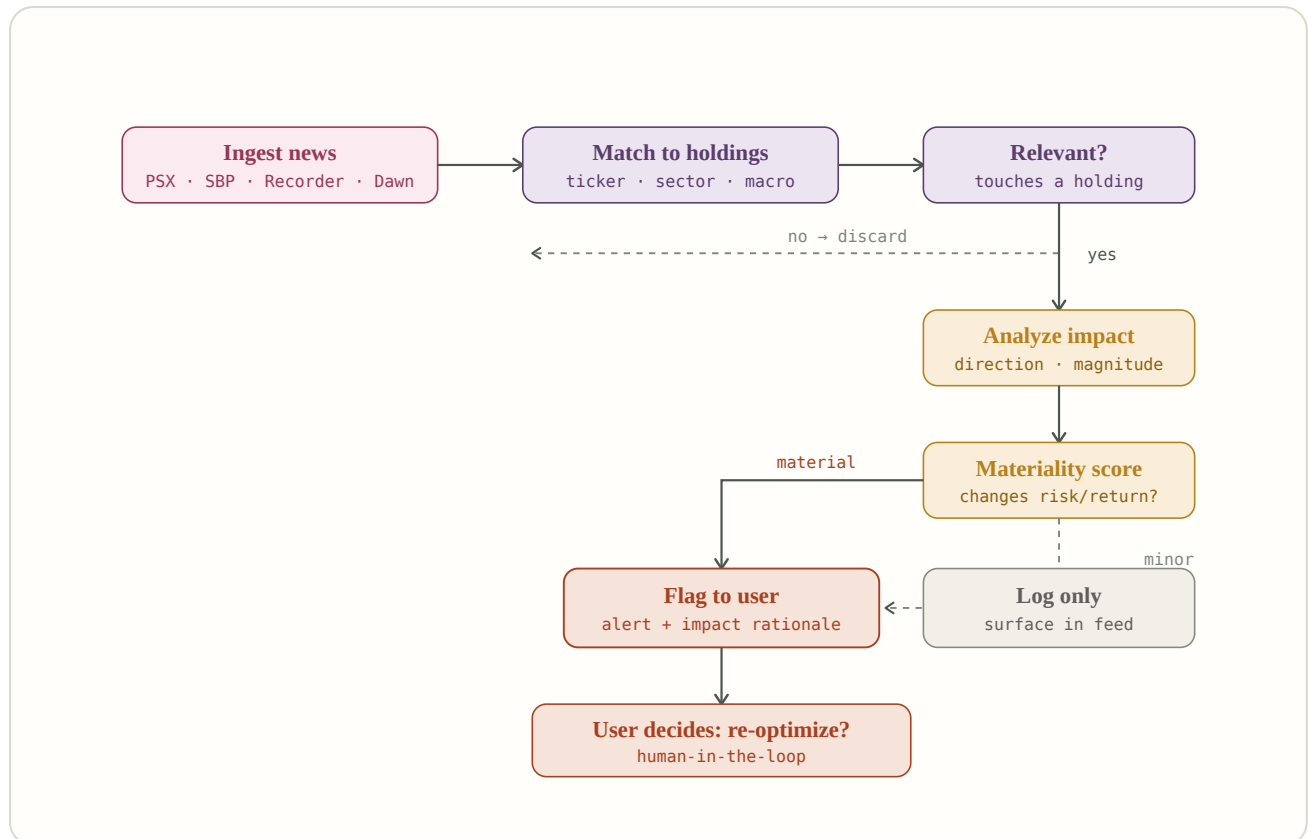


Figure 2 · **News-to-flag pipeline** – relevance filter, impact analysis, materiality gate, and the human decision point

Three levels of relevance

LEVEL	TRIGGER	EXAMPLE
Direct	News names a ticker the user holds	"HBL announces record profits" → user holds HBL
Sector	News affects a sector the user is exposed to	SBP raises rates → user holds bank stocks
Macro	Portfolio-wide effect	PKR depreciates 5% → affects everything

FIXED-INCOME DEPTH

For T-bills the agent encodes the inverse rate relationship: rates up → existing bond prices down, new issues yield more. An SBP rate cut prompts "your T-bills just appreciated"; a hike prompts "market value may dip, but reinvestment yield improves." Getting this relationship right is what makes the fixed-income guidance trustworthy.

07 Data layer

Each asset class gets a refresh strategy matched to how fast it actually moves and how its data is available. This keeps the system within free-tier limits.

ASSET CLASS	PRIMARY SOURCE	FALLBACK	REFRESH
Global stocks	yfinance	—	Daily EOD
PSX stocks	yfinance .KA	PSX DPS scrape	Daily EOD
Crypto	CoinGecko	cached value	30–60s cache
T-bills	SBP scrape	—	Event-driven
Commodities	yfinance + FRED	Alpha Vantage	Daily + on-demand news

KEY SIMPLIFICATION

yfinance carries PSX stocks via the .KA suffix (NBP.KA, PSO.KA, HBL.KA...), so it drives global stocks, PSX stocks, *and* commodities through one adapter. PSX scraping is demoted to a targeted fallback for freshness checks, missing tickers, and dividends — far less fragile than a scrape-first pipeline.

Refresh orchestration

- ◆ **Daily (after close):** yfinance EOD for stocks + commodities → recompute portfolio values + write a daily snapshot.
- ◆ **Frequent / on-load:** CoinGecko crypto, batched and cached 30–60s.
- ◆ **Event-driven:** T-bill scrape on new bond announcement or 3-month timer; rate-impact analysis on SBP rate news.
- ◆ **On-demand:** commodity news when the user asks; full portfolio refresh on request.

08 LLM routing

The agent runs free-first, routing each call to the provider best suited for it. High-stakes data privacy is handled by keeping real financial PII out of free tiers.

JOB	PROVIDER / MODEL	WHY
Reasoning & analysis	Gemini 2.5 Flash	1,500 req/day, 1M tokens/min — handles large news + portfolio context
User-facing chat	Groq (Llama 3.3 70B)	Fastest inference; low latency for conversation
News sentiment (bulk)	Gemini Flash-Lite / Groq 8B	Cheap, fast, high volume
Rate-limit fallback	OpenRouter free slot	Re-route when a primary provider caps out

PRIVACY GUARDRAIL

Gemini's free tier may use prompts for training. The agent reasons over *abstracted* data (ticker + weight, not real account IDs or balances) — consistent with the platform's privacy-by-architecture principle. Verify current free-tier quotas before building; providers change them often.

09 Feature requirements

FEATURE	REQUIREMENT	PRIORITY
Onboarding profiler	Chat/form captures risk, horizon, capital, constraints; classifies investor mode	P0
Portfolio optimizer	MPT via PyPortfolioOpt, SBP rate as risk-free, sector concentration limits	P0
Suggestion + rationale	Allocation chart, expected return/risk, plain-language justification	P0
Holdings confirmation	User locks actual positions; portfolio moves to tracked state	P0
Performance dashboard	Value, P&L, per-class breakdown, time-series chart from daily snapshots	P0
Personalized news feed	News filtered to holdings, tagged with impact direction	P0
Materiality flagging	Background job scores events, alerts on material ones with rationale	P0
Re-optimization	User-triggered; agent factors in the flagging reason	P0
Rate-impact engine	SBP rate news → T-bill yield/price recompute + explanation	P1
Technical indicators	RSI, MACD, MAs on daily data for active mode	P1
Drift detection	Flag when allocation drifts from target; suggest rebalance	P2
Mutual funds (MUFAP)	Include fund data — matches real retail behavior	P2

10 Open decisions

Four choices still shape the build. Recommendations noted, but these are owner calls.

T-bill valuation depth

Proper bond pricing (discount future cash flows at current yield) vs. simplified directional impact.

Lean simplified for MVP; proper pricing as a P1 upgrade.

Daily portfolio snapshots

Store a daily value snapshot per portfolio to power the performance-over-time chart.

Yes — small effort now, enables the performance chart.

Single vs. multi-portfolio

One portfolio per user, or separate long-term and short-term portfolios.

Single for MVP; multi adds state complexity.

Crypto weighting

Crypto is legal in Pakistan and popular with retail; the open question is how much weight it gets in the default universe, not whether to include it.

Include by default; cap exposure via risk constraints.

11 MVP scope & cuts

IN SCOPE	OUT OF SCOPE (MVP)
Onboarding → suggested portfolio	cut Live trade execution — show plan only
Confirm holdings → tracking	cut Intraday / tick data
Daily news scrape → match → materiality flag	cut Auto-rebalancing on news
Performance dashboard + snapshots	cut Full PSX universe — start with liquid KSE-100 subset
Chat: portfolio + news-impact questions	cut Multi-portfolio per user

REFERENCE WALKTHROUGH

Fill form → get suggested portfolio → confirm → a material news event (e.g. an SBP rate decision) → agent flags it with impact analysis → user triggers re-optimize → agent shows the adjusted portfolio. This single loop exercises every core part of the product end to end.

WHY START WITH LIQUID NAMES

Thinly-traded PSX stocks have gappy history → unstable covariance matrices → garbage optimization. Starting with the most liquid 30–50 KSE-100 names gives clean correlations, stable optimization, and reliable output. Quality over coverage.

12 Risks & guardrails

RISK	GUARDRAIL
Scraping fragility	Prefer structured/RSS sources (PSX announcements); yfinance-primary cuts scrape surface area
Free-tier rate limits	Aggressive caching, batched calls, per-class refresh cadence, OpenRouter fallback
Data freshness (yfinance .KA)	Freshness check; fall back to PSX DPS for the current close when stale
Financial-advice liability	Frame outputs as "suggested allocation based on MPT," with disclaimers; agent never auto-acts
Local price accuracy (gold)	International price $\neq$ local PKR price; scrape local rate or convert via PKR/USD FX
LLM prompt privacy	Reason over abstracted data; keep real PII out of free tiers

NON-NEGOTIABLE

The agent suggests; the user decides. No automatic rebalancing, no trade execution without explicit confirmation. This is both a trust requirement and a compliance one.